

OVERVIEW

NOSQL DATABASES - RANKING

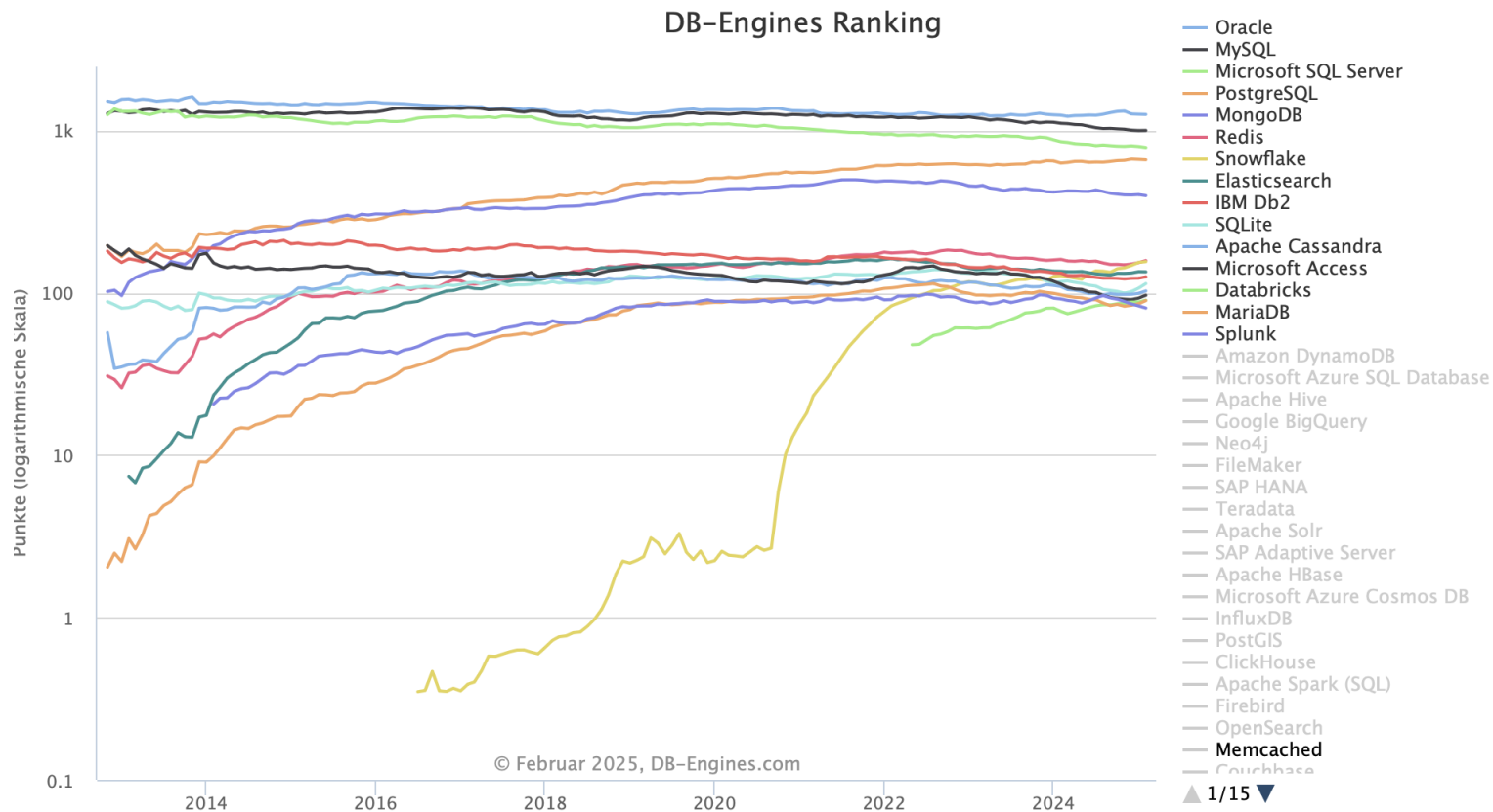
RANKING OF ALL DATABASES

424 Systeme im Ranking, Februar 2025

Rang			DBMS	Datenbankmodell	Punkte		
Feb 2025	Jan 2025	Feb 2024			Feb 2025	Jan 2025	Feb 2024
1.	1.	1.	Oracle +	Relational, Multi-Model ⓘ	1254,82	-3,93	+13,38
2.	2.	2.	MySQL +	Relational, Multi-Model ⓘ	999,99	+1,84	-106,67
3.	3.	3.	Microsoft SQL Server	Relational, Multi-Model ⓘ	786,87	-11,69	-66,70
4.	4.	4.	PostgreSQL +	Relational, Multi-Model ⓘ	659,62	-3,79	+30,21
5.	5.	5.	MongoDB +	Document, Multi-Model ⓘ	396,63	-5,87	-23,73
6.	↑ 7.	6.	Redis +	Key-value, Multi-Model ⓘ	157,91	+4,55	-2,80
7.	↓ 6.	↑ 9.	Snowflake +	Relational	155,58	+1,68	+28,13
8.	8.	↓ 7.	Elasticsearch	Multi-Model ⓘ	134,63	-0,29	-1,11
9.	9.	↓ 8.	IBM Db2	Relational, Multi-Model ⓘ	125,44	+2,46	-6,79
10.	10.	10.	SQLite	Relational	113,82	+7,13	-3,47
11.	11.	↑ 12.	Apache Cassandra +	Wide column, Multi-Model ⓘ	102,58	+3,39	-6,69
12.	12.	↓ 11.	Microsoft Access	Relational	96,54	+3,84	-16,63
13.	13.	↑ 17.	Databricks +	Multi-Model ⓘ	90,03	+2,19	+13,13
14.	14.	↓ 13.	MariaDB +	Relational, Multi-Model ⓘ	89,50	+3,92	-7,73
15.	15.	↓ 14.	Splunk	Suchmaschine	80,56	-2,53	-11,09
16.	↑ 17.	↓ 15.	Amazon DynamoDB +	Multi-Model ⓘ	75,58	+2,58	-7,31
17.	↓ 16.	↓ 16.	Microsoft Azure SQL Database	Relational, Multi-Model ⓘ	72,76	-1,02	-6,80
18.	18.	18.	Apache Hive	Relational	62,49	+5,61	-3,32
19.	19.	19.	Google BigQuery +	Relational	54,89	+1,85	-8,74
20.	20.	↑ 22.	Neo4j +	Graph	45,34	+1,65	-1,27
21.	21.	21.	FileMaker	Relational	40,22	-1,28	-10,26
22.	↑ 23.	↑ 23.	SAP HANA +	Relational, Multi-Model ⓘ	35,22	-0,84	-10,00

RANKING OF ALL DATABASES

Februar 2025



Klicken Sie auf ein System in der Legende um seine Trendlinie ein- oder auszublenden

COURSE PROJECT TOPICS/DATABASES

- **Column Oriented DB:**

- Cassandra
- HBase

- **Key Value DB:**

- Hazelcast (+Multi-Model)
- Arango DB (+Multi-Model)
- Redis

- **Document Oriented DB:**

- CouchbaseDB
- Mongo DB

- **Graph DB:**

- Neo4j

- **Additional Topic** – please, contact the instructor for approval

RANKING WIDE COLUMN STORE

□ include secondary database models

13 systems in ranking, February 2025

Rank			DBMS	Database Model	Score		
Feb 2025	Jan 2025	Feb 2024			Feb 2025	Jan 2025	Feb 2024
1.	1.	1.	Apache Cassandra 	Wide column, Multi-model 	102.58	+3.39	-6.69
2.	2.	2.	Apache HBase	Wide column	24.48	-0.27	-9.47
3.	3.	3.	Microsoft Azure Cosmos DB 	Multi-model 	22.13	-0.83	-9.86
4.	4.	4.	Datastax Enterprise 	Wide column, Multi-model 	4.22	-0.26	-3.43
5.	5.	5.	ScyllaDB 	Wide column, Multi-model 	3.65	-0.01	-2.51
6.	6.	6.	Microsoft Azure Table Storage	Wide column	2.73	-0.09	-2.92
7.	7.	 8.	Google Cloud Bigtable	Multi-model 	2.70	-0.09	-1.46
8.	8.	 7.	Apache Accumulo	Wide column	2.62	-0.17	-1.59
9.	9.	 10.	Amazon Keyspaces	Wide column	0.97	-0.01	+0.12
10.	10.	 9.	HPE Ezmeral Data Fabric	Multi-model 	0.76	-0.03	-0.45
11.	11.	11.	Elassandra	Wide column, Multi-model 	0.29	0.00	-0.01
12.	12.	12.	Alibaba Cloud Table Store	Wide column	0.21	+0.00	+0.02
13.	13.	13.	SWC-DB	Wide column, Multi-model 	0.01	0.00	0.00

DB-Engines(2025): <https://db-engines.com/en/ranking/wide+column+store> ; retrieved on 03. Feb. 2025

RANKING KEY VALUE STORES

□ sekundäre Datenbankmodelle berücksichtigen

71 Systeme im Ranking, Februar 2025

Rang			DBMS	Datenbankmodell	Punkte		
Feb 2025	Jan 2025	Feb 2024			Feb 2025	Jan 2025	Feb 2024
1.	1.	1.	Redis +	Key-value, Multi-Model ⓘ	157,91	+4,55	-2,80
2.	2.	2.	Amazon DynamoDB +	Multi-Model ⓘ	75,58	+2,58	-7,31
3.	3.	3.	Microsoft Azure Cosmos DB +	Multi-Model ⓘ	22,13	-0,83	-9,86
4.	4.	4.	Memcached	Key-value	16,69	+0,09	-3,72
5.	5.	↑ 6.	etcd	Key-value	7,06	-0,02	-1,08
6.	6.	↓ 5.	Hazelcast	Key-value, Multi-Model ⓘ	5,52	-0,07	-2,66
7.	7.	7.	Aerospike +	Multi-Model ⓘ	5,09	+0,05	-2,00
8.	8.	8.	Ehcache	Key-value	4,32	-0,07	-1,58
9.	9.	9.	Riak KV	Key-value	3,33	-0,14	-1,90
10.	10.	↑ 15.	Oracle NoSQL	Multi-Model ⓘ	3,29	+0,05	-0,50
11.	↑ 15.	↓ 10.	RocksDB	Key-value	3,06	+0,33	-1,46
12.	12.	12.	ArangoDB +	Multi-Model ⓘ	2,91	-0,01	-1,43
13.	↓ 11.	↓ 11.	Apache Ignite	Multi-Model ⓘ	2,89	-0,05	-1,52
14.	14.	↑ 16.	OrientDB	Multi-Model ⓘ	2,73	-0,06	-0,92
15.	↓ 13.	↓ 13.	Google Cloud Bigtable	Multi-Model ⓘ	2,70	-0,09	-1,46
16.	↑ 17.	↑ 19.	GemFire	Key-value, Multi-Model ⓘ	2,43	-0,01	-0,97
17.	↓ 16.	↓ 14.	InterSystems IRIS +	Multi-Model ⓘ	2,35	-0,21	-1,62
18.	18.	18.	LevelDB	Key-value	2,12	+0,02	-1,39

RANKING DOCUMENT STORES

□ sekundäre Datenbankmodelle berücksichtigen

58 Systeme im Ranking, Februar 2025

Rang			DBMS	Datenbankmodell	Punkte		
Feb 2025	Jan 2025	Feb 2024			Feb 2025	Jan 2025	Feb 2024
1.	1.	1.	MongoDB +	Document, Multi-Model ⓘ	396,63	-5,87	-23,73
2.	2.	↑ 3.	Databricks +	Multi-Model ⓘ	90,03	+2,19	+13,13
3.	3.	↓ 2.	Amazon DynamoDB +	Multi-Model ⓘ	75,58	+2,58	-7,31
4.	4.	4.	Microsoft Azure Cosmos DB +	Multi-Model ⓘ	22,13	-0,83	-9,86
5.	5.	5.	Couchbase +	Multi-Model ⓘ	15,63	-0,38	-4,85
6.	6.	6.	Firebase Realtime Database	Document	13,32	+0,21	-3,02
7.	7.	7.	CouchDB	Document, Multi-Model ⓘ	7,81	+0,03	-4,86
8.	↑ 9.	8.	Google Cloud Firestore	Document	6,92	+0,19	-3,95
9.	↓ 8.	9.	Realm	Document	6,74	-0,25	-1,22
10.	10.	↑ 11.	Aerospike +	Multi-Model ⓘ	5,09	+0,05	-2,00
11.	11.	↓ 10.	MarkLogic	Multi-Model ⓘ	4,16	+0,21	-3,33
12.	12.	↑ 13.	Google Cloud Datastore	Document	3,69	-0,11	-1,60
13.	↑ 14.	↑ 17.	Oracle NoSQL	Multi-Model ⓘ	3,29	+0,05	-0,50
14.	↓ 13.	↓ 12.	Virtuoso +	Multi-Model ⓘ	3,14	-0,18	-2,18
15.	15.	↓ 14.	ArangoDB +	Multi-Model ⓘ	2,91	-0,01	-1,43
16.	16.	↑ 18.	OrientDB	Multi-Model ⓘ	2,73	-0,06	-0,92
17.	↑ 19.	↓ 16.	RavenDB	Document, Multi-Model ⓘ	2,53	+0,03	-1,41
18.	↓ 17.	↑ 21.	IBM Cloudant	Document	2,52	-0,08	-0,59
19.	↓ 18.	↓ 15.	InterSystems IRIS +	Multi-Model ⓘ	2,35	-0,21	-1,62
20.	20.	↓ 19.	RethinkDB	Document, Multi-Model ⓘ	2,30	-0,07	-0,93
21.	21.	↑ 22.	PouchDB	Document	1,87	-0,07	-1,01

DB-Engines(2025): <https://db-engines.com/de/ranking/document+store> ; retrieved on 03. Feb. 2025

RANKING GRAPH DBs

☐ sekundäre Datenbankmodelle berücksichtigen

43 Systeme im Ranking, Februar 2025

Rang			DBMS	Datenbankmodell	Punkte		
Feb 2025	Jan 2025	Feb 2024			Feb 2025	Jan 2025	Feb 2024
1.	1.	1.	Neo4j	Graph	45,34	+1,65	-1,27
2.	2.	2.	Microsoft Azure Cosmos DB	Multi-Model	22,13	-0,83	-9,86
3.	3.	3.	Aerospike	Multi-Model	5,09	+0,05	-2,00
4.	4.	4.	Virtuoso	Multi-Model	3,14	-0,18	-2,18
5.	5.	5.	ArangoDB	Multi-Model	2,91	-0,01	-1,43
6.	6.	6.	OrientDB	Multi-Model	2,73	-0,06	-0,92
7.	7.	9.	GraphDB	Multi-Model	2,63	-0,09	-0,12
8.	8.	7.	Memgraph	Graph	2,54	-0,08	-0,63
9.	9.	8.	Amazon Neptune	Multi-Model	2,12	-0,07	-0,81
10.	10.	11.	JanusGraph	Graph	1,69	-0,03	-0,61
11.	11.	12.	Stardog	Multi-Model	1,61	-0,07	-0,52
12.	12.	10.	NebulaGraph	Graph	1,58	-0,08	-0,96
13.	13.	14.	Fauna	Multi-Model	1,49	+0,07	-0,30
14.	14.	13.	TigerGraph	Graph	1,45	+0,03	-0,52
15.	15.	15.	Dgraph	Graph	1,20	-0,05	-0,40
16.	16.	16.	Giraph	Graph	1,07	-0,01	-0,23

OVERVIEW – LIST OF NoSQL DB

- Cloud Firestore – Cloud System, Document Store
- CouchBase – Document Store
- CouchDB – Document Store
- RavenDB – Document Store
- RethinkDB – Document Store
- MongoDB – Document Store
- BigChainDB - Blockchain technology with Document Store
- ElasticSearch – Search Engine with Document Store (MongoDB, RethinkDB)
- ArangoDB – Multimodel (Document Store, Key-Value, Graph)
- Riak – Key-Value DB
- Redis – Key-Value DB
- Aerospike – In Memory Key-Value DB
- OrientDB – Document Store properties with Graph DB
- Neo4j – Graph DB
- Cassandra – Column Oriented DB
- HBase – Column Oriented DB (Hadoop)

INTRODUCTION TO NoSQL

NoSQL DEFINITION

No standard definition of NoSQL!

No SQL

and

Not only SQL

... type of database management system that's
useful for very large sets of distributed data ...

INTRODUCTION NoSQL

History (Edlich et al., 2011, 1.1 Historie):

- 1979** Ken Thompson developed a Key/Hash database, called DBM.
 - 1980s** Development the pioneer of NoSQL systems like Lotus Notes, BerkeleyDB
 - 1998** For the first time the term of NoSQL appeared in a database by Carlo Strozzi (IBM).
 - 2000** Along with the development of Web 2.0 attempted to process big data.
 - 2004** Development of BigTable database and NoSQL approach like Map/Reduce by Google (Pioneer for NoSQL)
Other companies as Yahoo, Amazon and Facebook etc. followed
 - 2006-2009** Development of classical NoSQL systems like CouchDB, Cassandra etc.
- => **Mai 2009** The term NoSQL in weblog of Eric Evans

NOSQL PROPERTIES

- Non-relational data model
- Vertical and horizontal scaling
- Open Source
- Schema-less database structure
- Distributed architecture
- Simple data replication
- Simple API
- Eventually consistency and BASE, but not ACID
- etc.

TYPES OF NoSQL DATABASES

- There are four NoSQL database classification:
 - Document Store
 - Key-Value Databases
 - Column Oriented Databases
 - Graph Databases

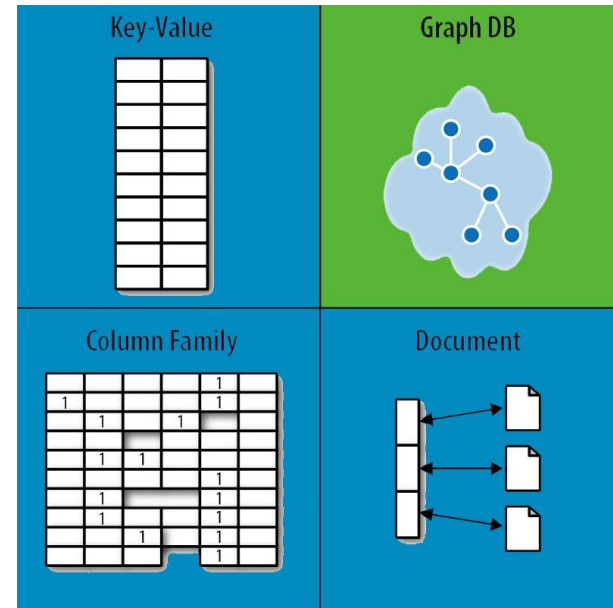


Figure 1: NoSQL Database types (Freecodecamp, 2019)

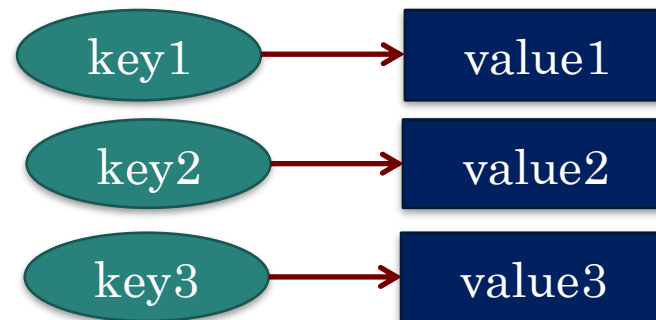
KEY VALUE DB (E.G. RIAK, REDIS, AMAZON DYNAMODB)

- Simple (!) Key and Value structure
- Primary Key access
- Great performance
- Easy to scale
- Features/functions in the program:
 - Only provide simple operations like get, put and delete
 - Data querying must be handled at the application layer

... like a relational table with two columns: Key/ID and Value

KEY VALUE DATABASES

- Can only store pairs of keys and values
- Can retrieve values when a key is known
- The key needs to be distinct (in a namespace or in the whole system)
- Has no pattern for the values
- Enables easy storage of any kind of data
 - Numbers
 - Strings
 - Binaries
 - Images
 - Short videos and more



OVERVIEW – LIST OF OPENSOURCE NoSQL DB

○ Key-Value DB

- Riak
 - <https://docs.riak.com/riak/>
 - basho.com/products/riak-kv
- Redis <https://redis.io/documentation>
- Hazelcast <https://hazelcast.org/documentation/>
- ArangoDB <https://www.arangodb.com/documentation/>

DOCUMENT ORIENTED DATABASES

- Everything is stored as a document
- Structured or semi-structured documents
- Each document has an unique key
- Document has internal notations:
 - Can be processed directly in applications
 - Mostly in JSON format
- Querying
 - support SQL statements
 - but most are queried with different languages

DOCUMENT ORIENTED DB

(E.G. COUCHDB, MONGODB)

- *also* Key and Value (document) structure
- Value (document) can be examined
- No scheme
- Similar but not the same document-structure
- Definition and control of document needs to be implemented

OVERVIEW – LIST OF OPENSOURCE NoSQL DB

○ Document Store

- CouchBase
<https://docs.couchbase.com/home/index.html>
- CouchDB <http://docs.couchdb.org/en/stable/>
- RethinkDB <https://rethinkdb.com/docs>
- MongoDB
<https://docs.mongodb.com/manual/core/document/>
- ElasticSearch (Search Engine)
<https://www.elastic.co/guide/en/elasticsearch/reference/current/index.html#>

COLUMN ORIENTED DATABASES

- Other name is wide column stores
- One row is identified by a key and can contain an arbitrary amount of columns.
- Stores data into collections of columns

COLUMN ORIENTED DATABASES

○ Data model

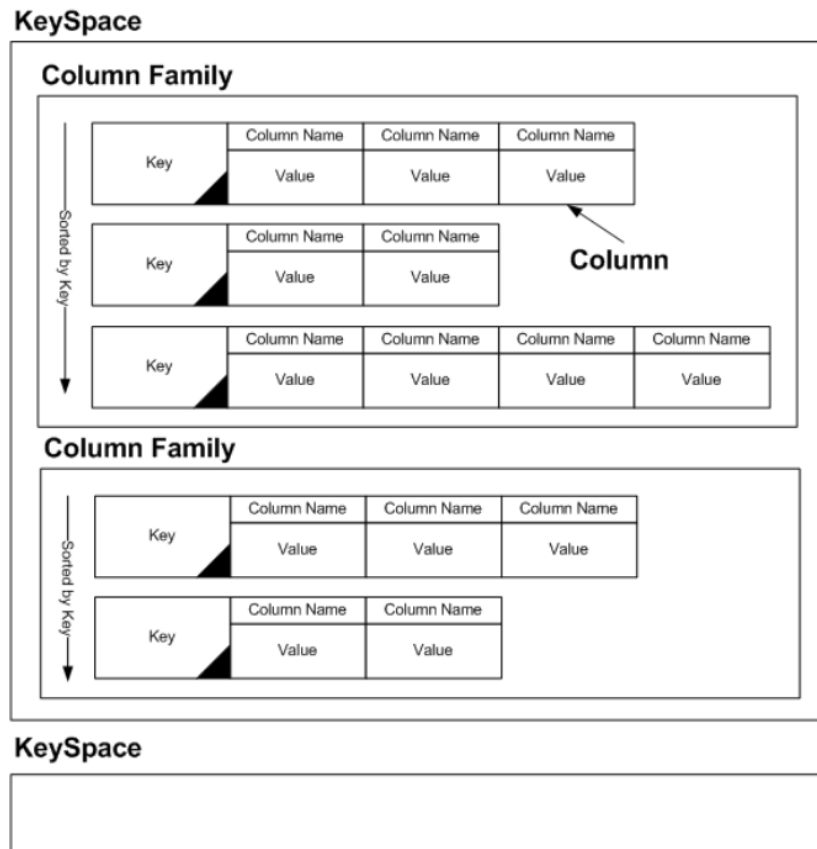


Figure 2: Data model of Column oriented Databases (Bruckner, 2012)

COLUMN ORIENTED DATABASES

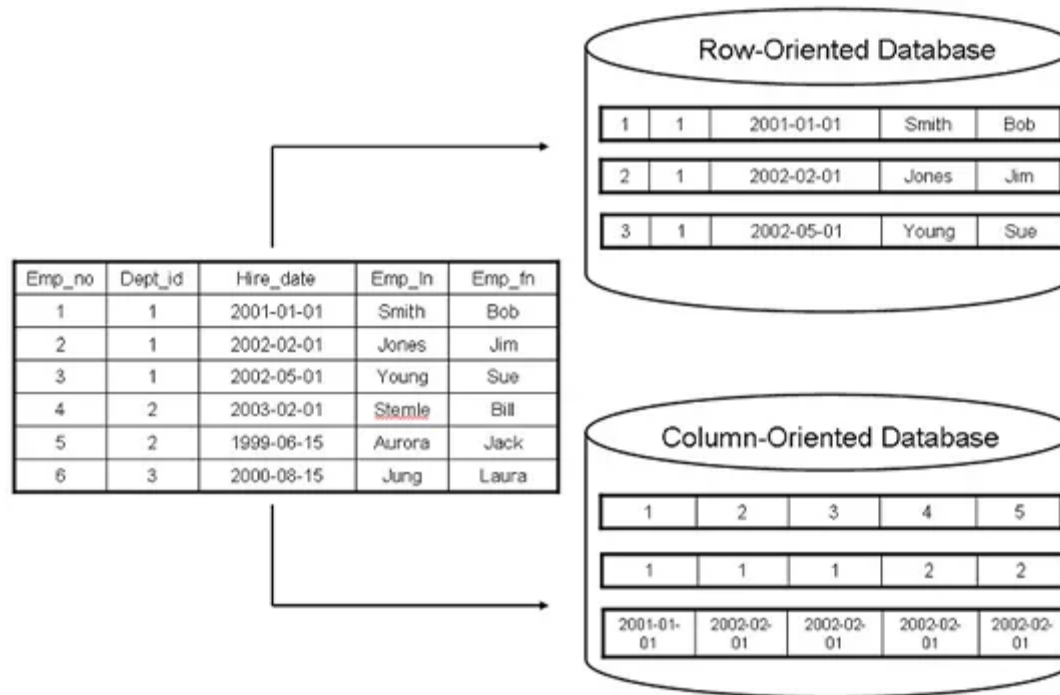


Figure 3: Column-Oriented Database Technologies (Db-best Technologies, 2012)

OVERVIEW – LIST OF OPENSOURCE NoSQL DB

○ Column Oriented DB

- Cassandra <https://cassandra.apache.org/doc/latest/>
- Hbase <https://hbase.apache.org/book.html>

GRAPH DATABASES

- represent data in graph structures as nodes and edges.
- Nodes are the entities
- Edges are the relationships between two node entities
- Attributes of nodes called properties (key-value pairs)
- Relationship can also contain properties

GRAPH DATABASES

○ Graph model

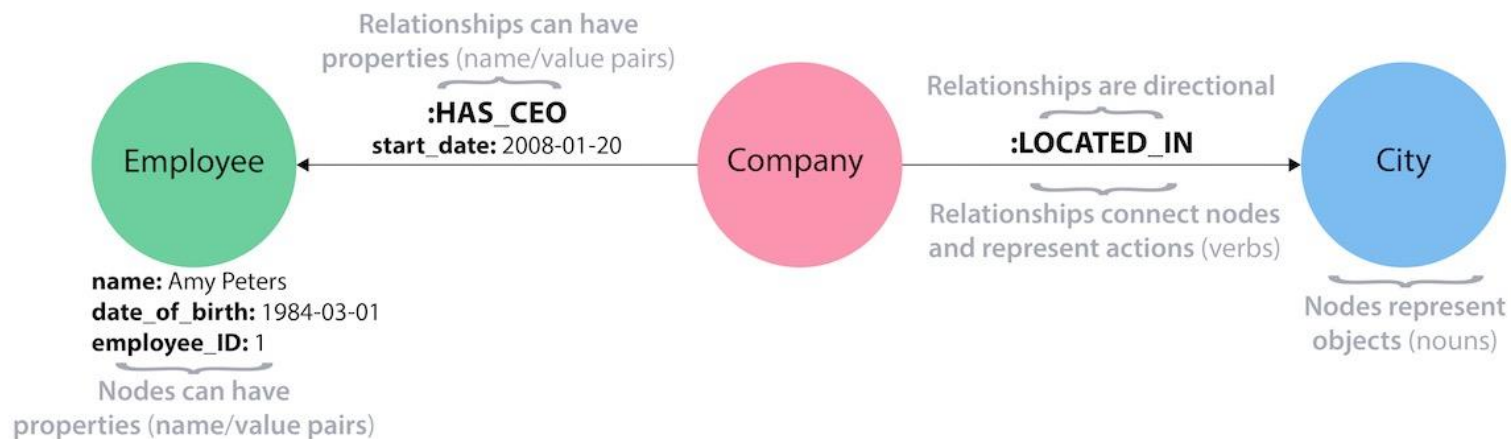


Figure 4: Graph database model (Neo4j, 2020)

OVERVIEW – LIST OF OPENSOURCE NoSQL DB

○ Graph DB

- Neo4j <https://neo4j.com/docs/>
- OrientDB <https://orientdb.org/docs/3.0.x/>
- ArangoDB <https://www.arangodb.com/documentation/>

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- Redis – Key-Value DB
- Aerospike – In Memory Key-Value DB
- OrientDB – Document Store properties with Graph DB
- Neo4j – Graph DB
- Cassandra – Column Oriented DB
- HBase – Column Oriented DB (Hadoop)

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